Time Series for Dummies

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ReLU

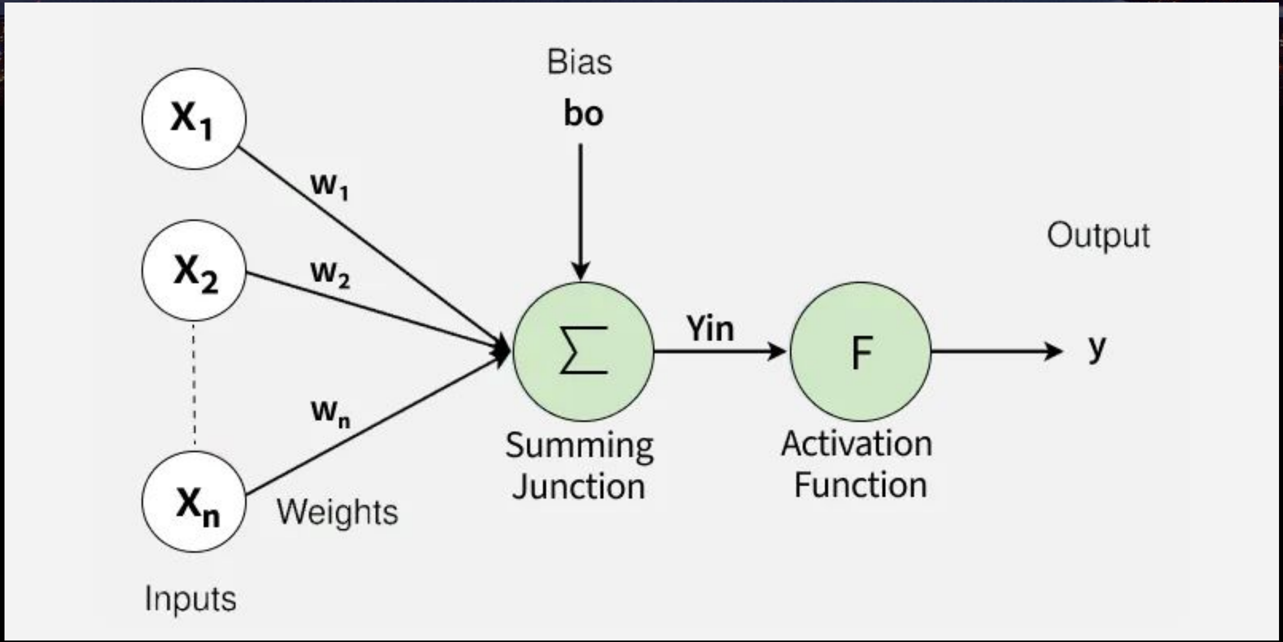
Structure for today

We are going to alternate between this presentation and the notebook I gave you

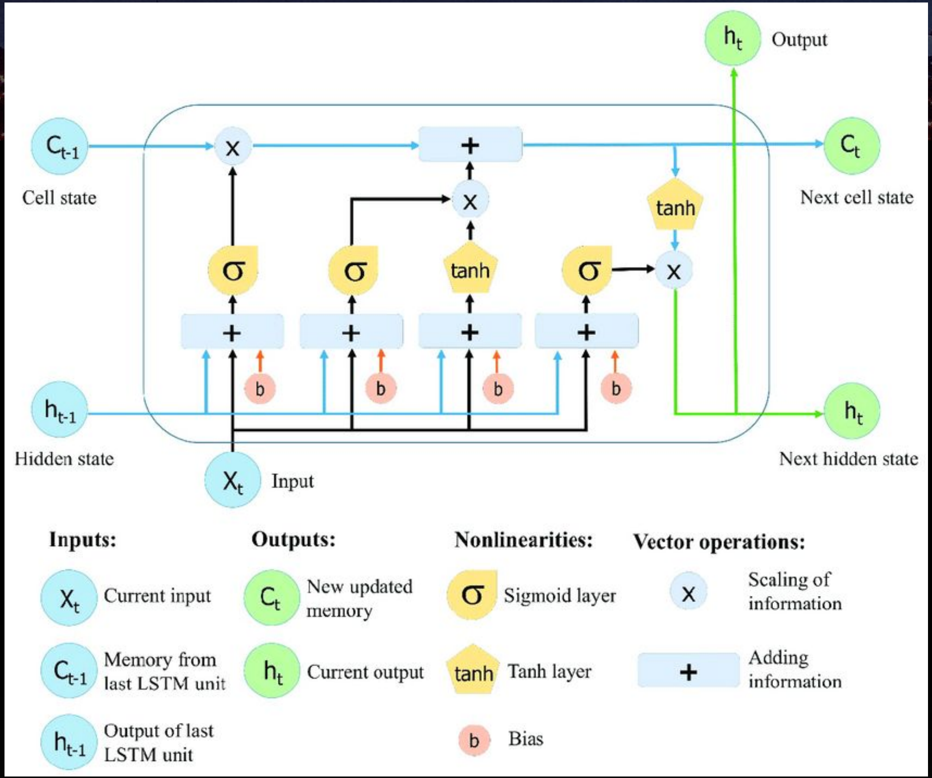
We are going to cover two things in parallel:

1. Build up to the **best models** for time series analysis (with Neural Networks at least)
2. Look at **important concepts** to consider when working with time series data

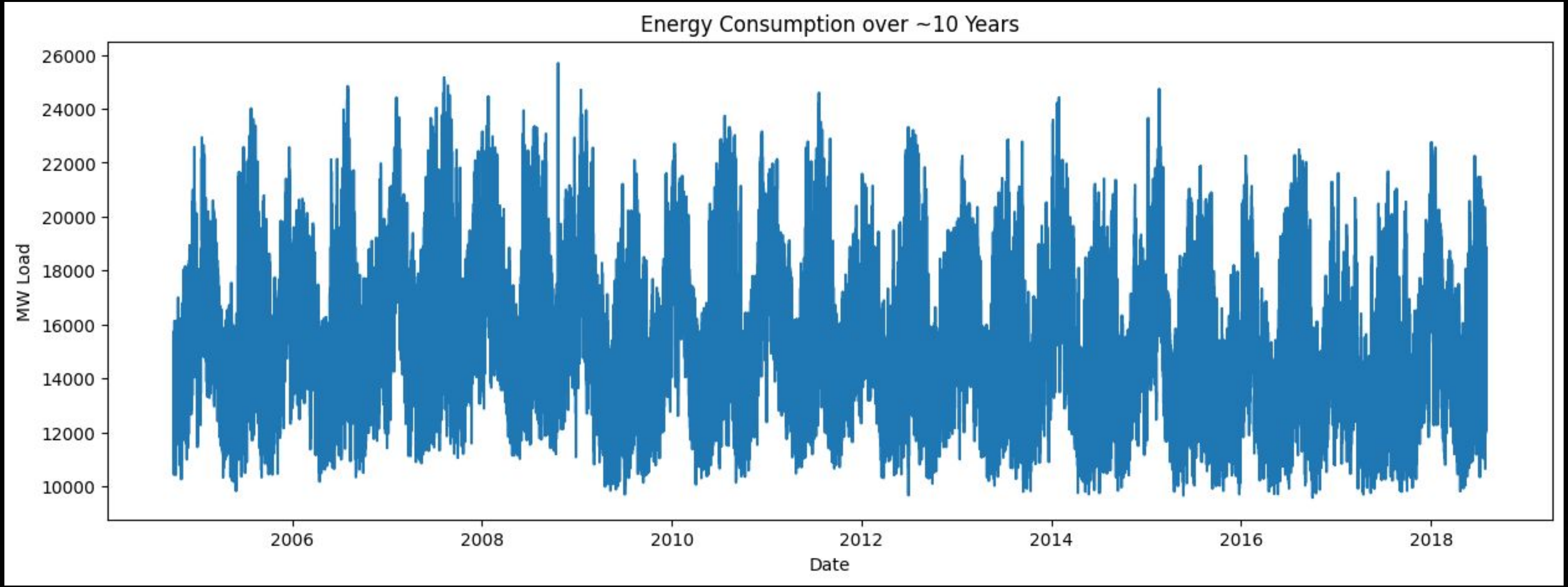
If we have a classic simple NN



What is all this and why do we need it?



Hourly Energy Consumption Dataset



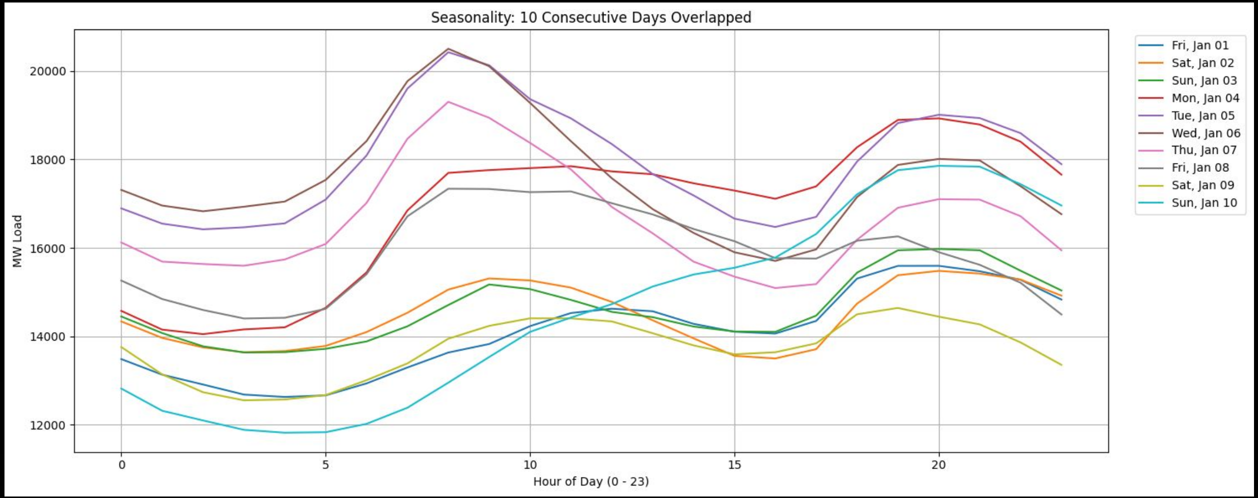
Concept #1: Seasonality vs Stationarity

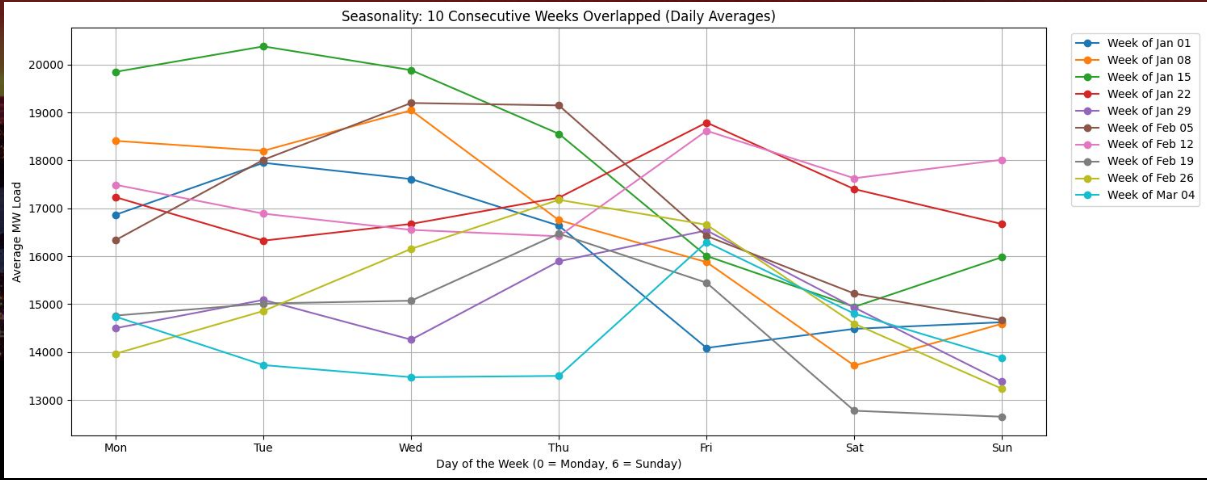
Linear Models usually work well if the data is **stationary:**

- Mean constant
- Variance constant
- Autocovariance constant

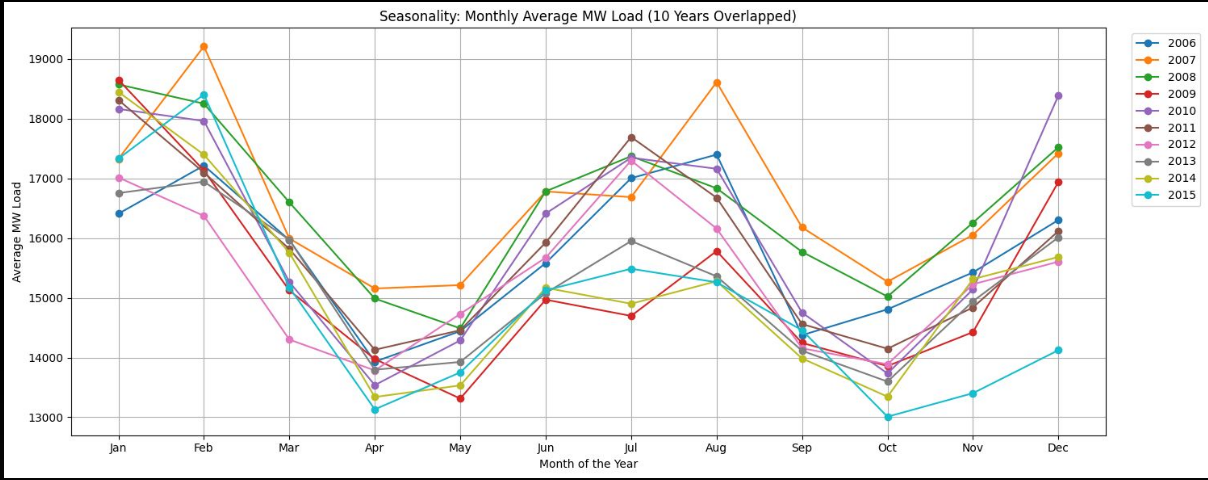
We say that a time series exhibits **seasonality** whenever there is a regular, periodic change in the mean of the series

Daily, Weekly and Yearly Seasonality





Seasonality: 10 Consecutive Weeks Overlapped (Daily Averages)
Average MW Load
Day of the Week (0 = Monday, 6 = Sunday)
Mon
Tue
Wed
Thu
Fri
Sat
Sun
13000
14000
15000
16000
17000
18000
19000
20000
Week of Jan 01
Week of Jan 08
Week of Jan 15
Week of Jan 22
Week of Jan 29
Week of Feb 05
Week of Feb 12
Week of Feb 19
Week of Feb 26
Week of Mar 04



Seasonality: Monthly Average MW Load (10 Years Overlapped)
Average MW Load
Month of the Year
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
13000
14000
15000
16000
17000
18000
19000
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015

Concept #2: Sequential Evaluation

Example method:

1. Evaluate the first hour based on the 24 previous hours
2. Update the new hours available with your prediction
3. Evaluate the second hour based on 23 actual values and 1 predicted
4. Update the new hours available with your second prediction
5. Evaluate the third hour based on the 22 actual values and 2 predicted

And so on!

Concept #3: Preventing Data leakage

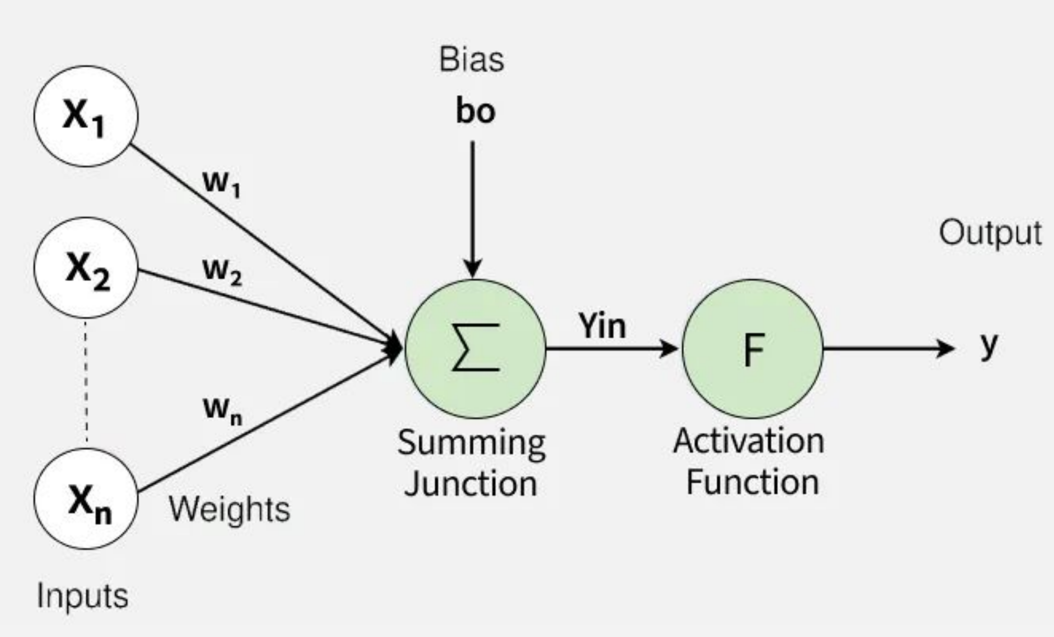
Data Leakage occurs when a model uses information during training that wouldn't be available at the time of prediction

In time series in particular, we need to make sure there is no information leaking **from the future**

- Missing values
- Variable transformations
- MinMax Scales

Let’s try to make the Feedforward NN work

Manually create 24 features representing the hours before the one we want to predict



Why it doesn’t work

- No concept of flow of time: features are indistinguishable to the model

- No ability to learn trends from a long time before

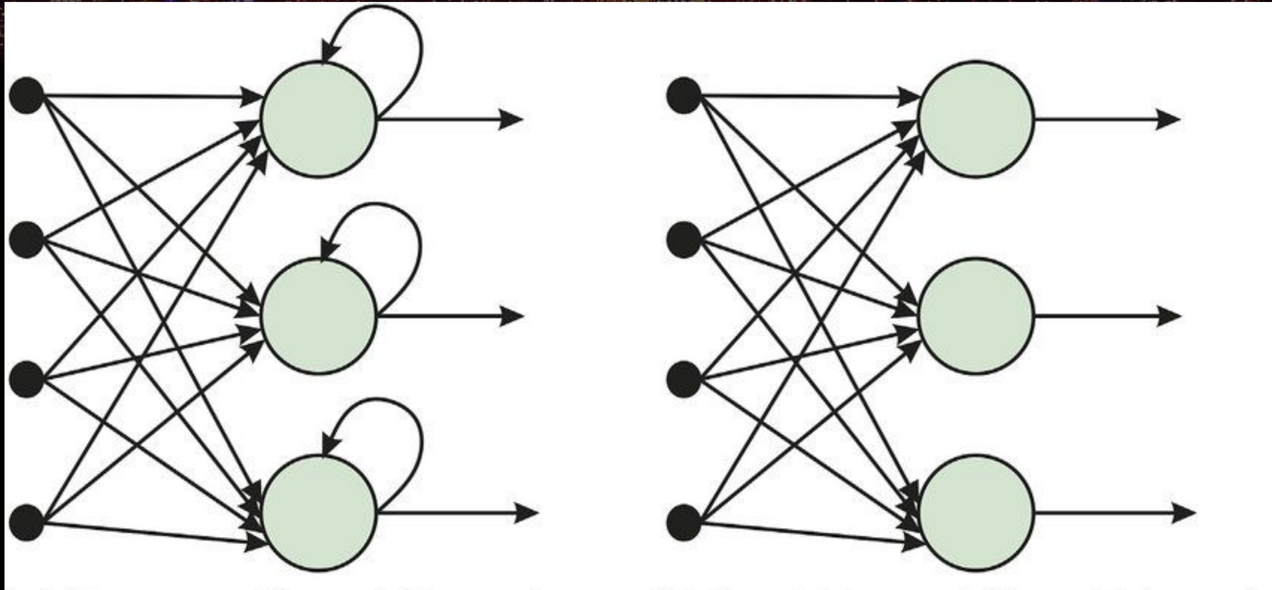
- Fails to be better than “guessing it’s equal to the previous time step”

Concept #4: lag features and more

Adding features engineered directly from your time series:
- Sine/cosine curve of seasonality
- Lag feature (value 24 hours before, 1 month before, 1 year before…)

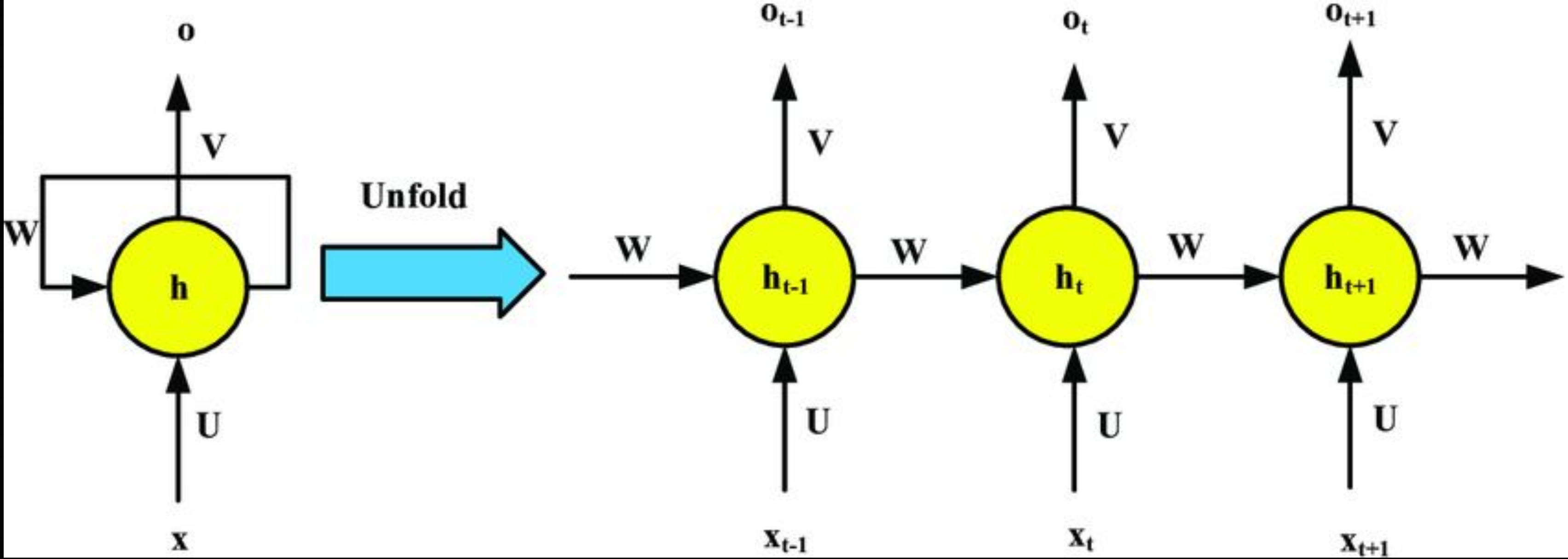
Think of things that would help you make a better prediction if you were to make it organically

Recurrent Neural Network

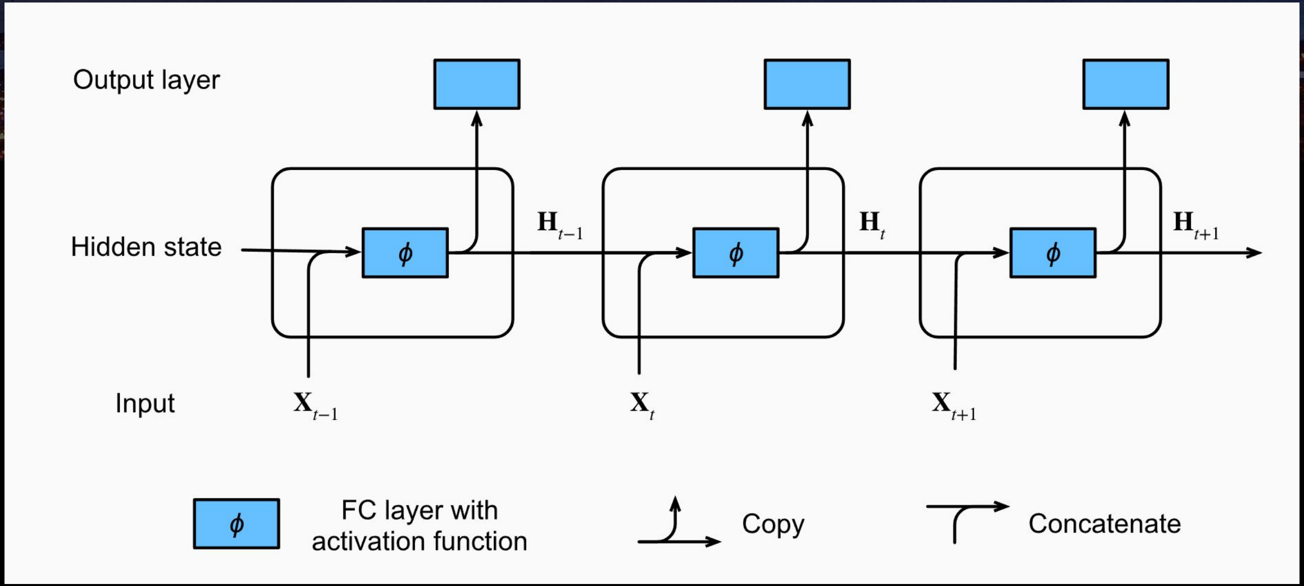


(a) Recurrent Neural Network (b) Feed-Forward Neural Network

RNN Architecture

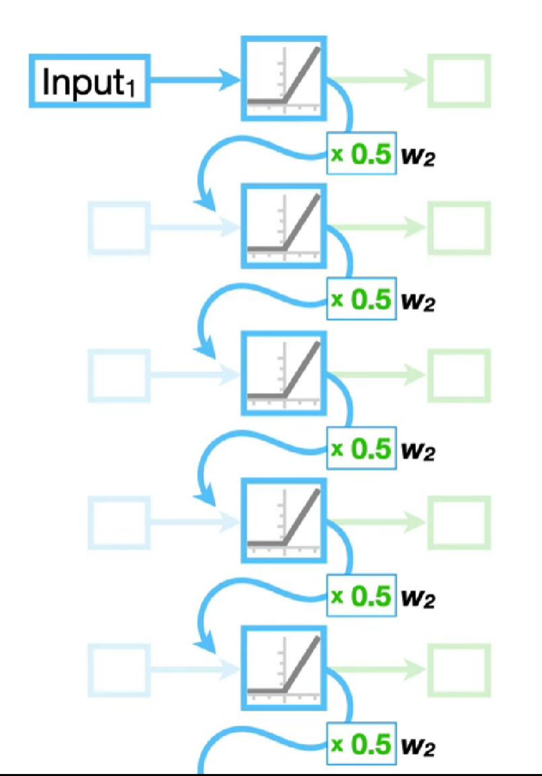


RNN Architecture



The **Hidden State** is what makes communication possible between adjacent time steps

Why it fails: Vanishing Gradient



With a weight from previous recurrence of 0.5, after just 10 steps back in time the influence of previous time steps is insignificant (<10^-3)

The model fails to learn any long term trend

Concept #5: Evaluation Metrics

Root Mean Square Error:

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

Mean Absolute Percentage Error

$$MAPE = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{A_i - F_i}{A_i} \right| \times 100$$

Concept #6: The Dumb Model

If you were to make your best guess without calculations, what would you do?

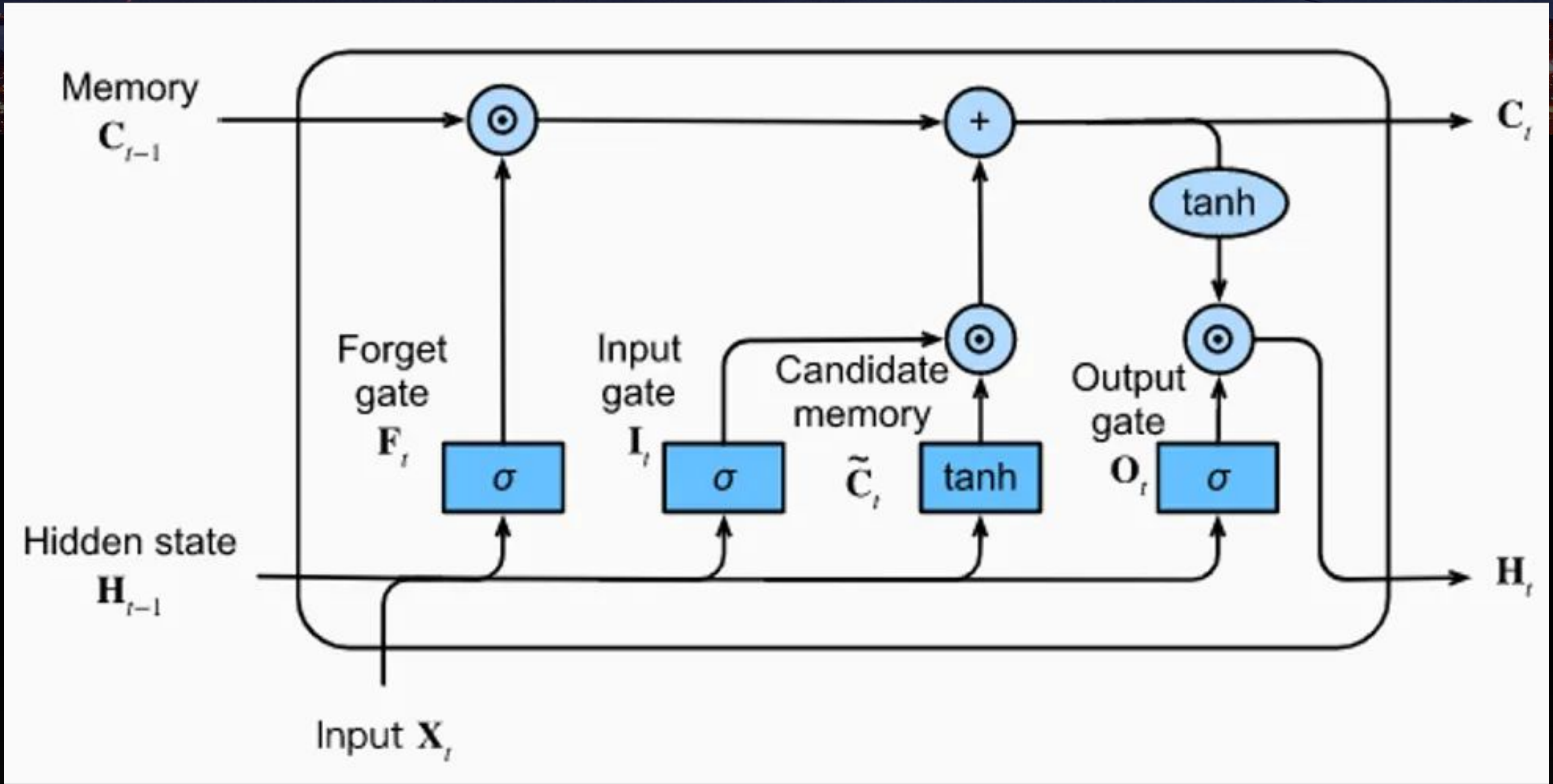
Concept #6: The Dumb Model

If you were to make your best guess without calculations, what would you do?

Compare with the model that just copies the values from

- previous day
- previous week
- previous month
- previous year

The LSTM architecture

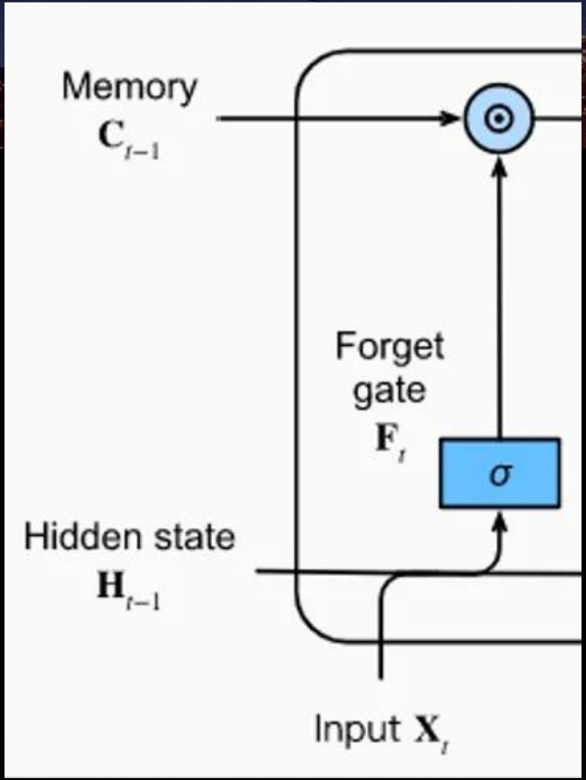


LSTM Components

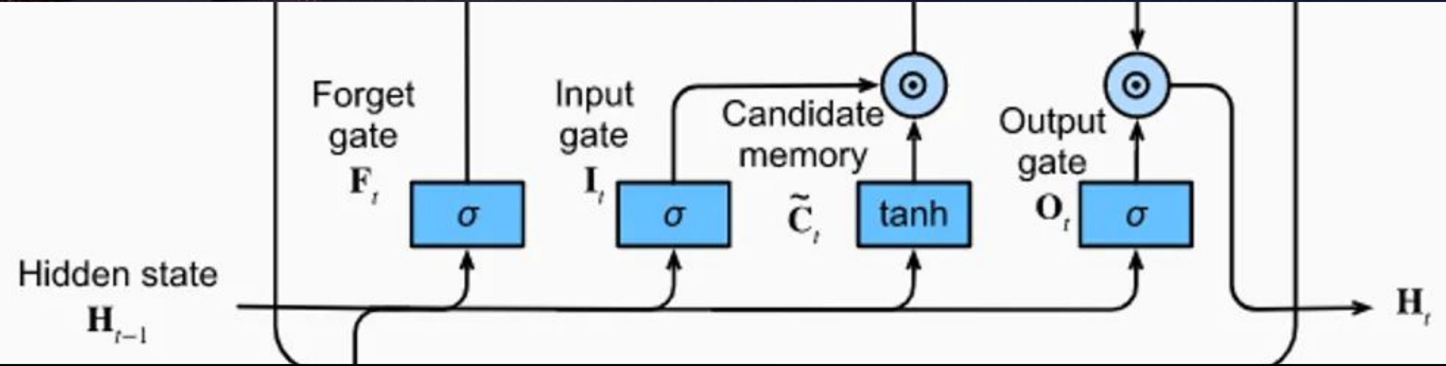
1. **Memory, or Cell State**: represents the long term memory of the model, passes through like a conveyor belt

2. **Hidden State:** the short term memory from the immediate previous steps

3. **Input:** the current value of the time series we are working with



The 3 LSTM Gates

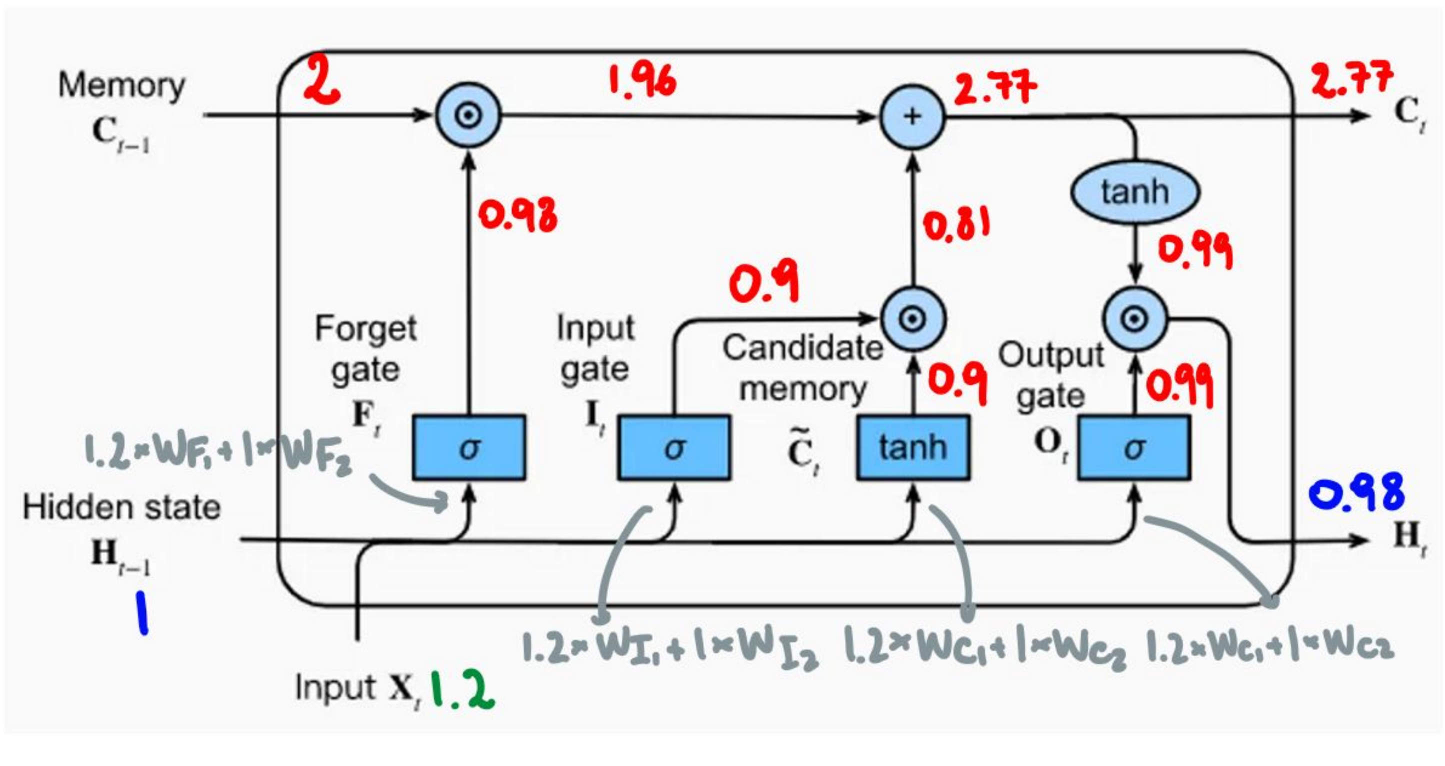


1. **Forget gate:** decides how much information from the long term memory is still relevant, given the input and short term memory

2. **Input Gate and Candidate Memory:** decide the new information to add to the long term memory

3. **Output Gate:** Update the short term memory

Time Series Data



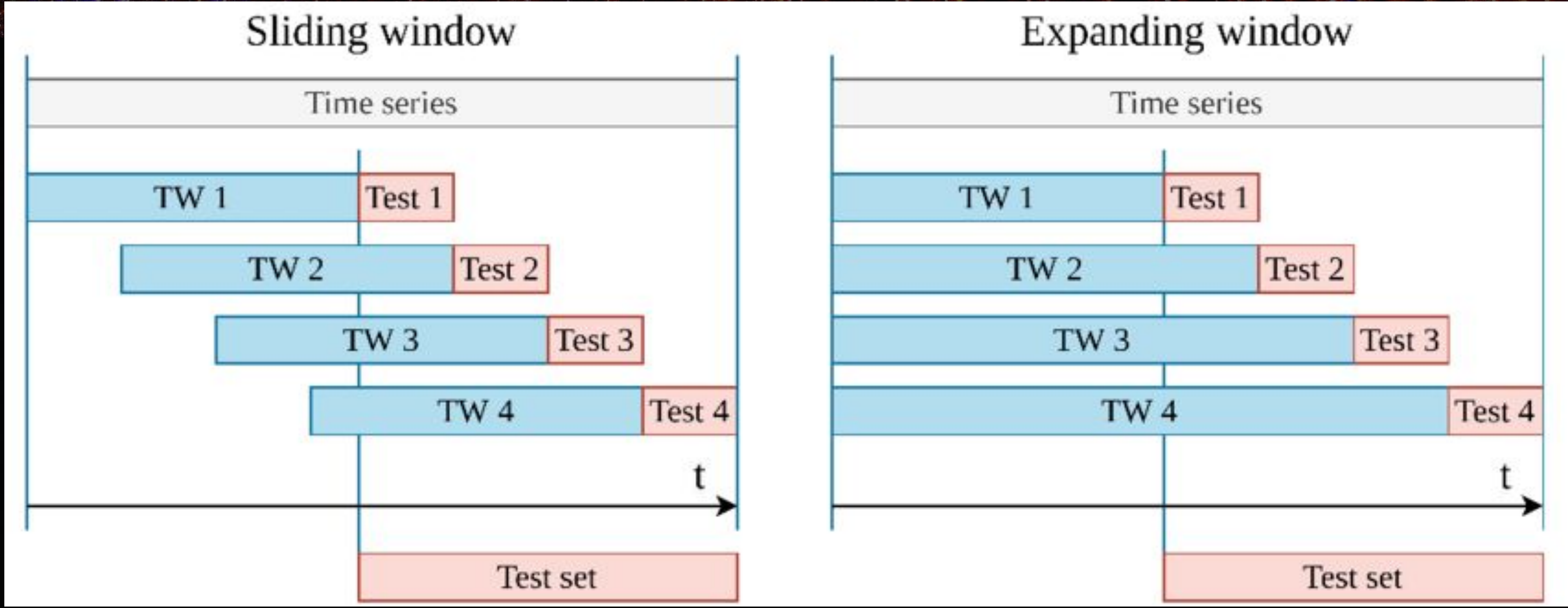
Concept #7: predicting chunks of time

Rather than predicting one hour at a time, let’s try to predict 24 hours all at once!

We need to structure the output of the model to be a vector of dimension 24 rather than 1

Allows for daily seasonality to be expressed in the output directly rather than the model storing all the complicated mechanics

Concept #8: Time-series Cross Validation



Feedback form

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